Kayra Bulbul

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Education

RMIT University Melbourne, VIC
Bachelor of Computer Science, Minors in data science and Enterprise Systems development *Mar. 2024 – Nov 2027*

Experience

Peer Mentor July 2026 – Present
RMIT University *Melbourne, VIC*

- Mentor students in **Algorithms & Analysis** and **Further Programming**, helping develop understanding of data structures, algorithms, OOP and debugging techniques.
- Guide students through problem-solving approaches and debugging strategies rather than providing direct solutions, strengthening their ability to independently reason about technical problems.

Projects

Trail | *Rust, Ratatui, SQLite, MCP* September 2026 – Present

- Built a **cross-platform terminal UI** in Rust with Ratatui and Crossterm that keeps a shared log of where developers and their AI coding agents left off on each project, for Linux, macOS and Windows.
- Implemented a **Model Context Protocol (MCP) server** from scratch over JSON-RPC on stdio, supporting both legacy and current spec revisions, so agents such as Claude Code and Codex can read and write project history as tools.
- Designed a **scriptable CLI** with clap and JSON output on a shared core layer used by the MCP server. Each update records the **git branch and commit** it was written on, and a read-only git view lets you browse branches, commits and diffs.
- Built a **SQLite storage layer** with backward-compatible schema migrations and an in-app self-updater that verifies GitHub Releases with **SHA-256 checksums**. Over 120 tests run on Linux, macOS and Windows in GitHub Actions CI.

Distributed Chat System | *Go, WebSockets, Redis, Docker, Prometheus/Grafana* August 2026 – September 2026

- Designed and built a **distributed real-time chat system** using WebSockets, supporting persistent client connections and communication across multiple server instances.
- Implemented **horizontal scaling** and **load balancing**, with Redis Pub/Sub enabling message propagation between users connected to different application servers.
- Engineered the system with an **infrastructure-focused architecture**, including containerised services with Docker, structured logging, metrics and observability, and load testing to evaluate behaviour under concurrent traffic.

Overworld Atlas | *React, TypeScript, Go, Docker, PostgreSQL* July 2026 – Present

- Built a **Go backend** service for a Minecraft server I play on with my mates, exposing APIs for server status, authentication and application functionality.
- Implemented the **Minecraft Server List Ping protocol** from scratch, including VarInt encoding, TCP packet construction, handshaking and status-response decoding.
- Added configurable **timeouts, TTL-based caching, CORS, structured logging and HTTP middleware**, with unit and integration tests covering protocol and API behaviour.

Achievements

Winner – MelbourneHack 2026 | *Python, FastAPI, PostgreSQL, ReactTS, Gemini* August 2026

- Built an accessibility platform in a four-person team that transforms web content based on customisable accessibility needs; led backend and database development using FastAPI and PostgreSQL, integrating page transformation, persistence and frontend services.

Technical Skills

Languages: Go, Python, Java, C/C++, JavaScript/TypeScript, Rust, SQL
Frameworks: React, Node.js, Express, FastAPI, Ratatui, Chi
Infrastructure & Tools: Git, Linux, Docker, Prometheus, Neovim, AWS
Data: PostgreSQL, MS SQL Server, SQLite, MongoDB, Redis